DIPLOMA PROGRAMME PRACTICE · SL · SET B

Biology / Paper 1B

Question booklet · 25 marks · 90 minutes across 1A and 1B

Show working, units and suitable precision. Use a calculator. Use separate answer paper for additional working and diagrams.

QUESTION 1 [6]

Comparing habitat surveys

Equal areas of two marshes are surveyed with identical effort. All recorded individuals belong to the three listed native species. A second survey of marsh Y adds one previously overlooked rare species.

Marsh	Species J	Species K	Species L
X	30	30	30
Y	75	10	5

(a) Compare initial species richness and evenness in the two marshes. **[2]**

(b) Calculate the percentage of recorded individuals in marsh Y that belong to J. Explain what the second survey reveals about observed richness. **[2]**

(c) Propose a sampling change to compare the marshes more reliably and state how it improves the evidence. **[2]**

QUESTION 2 [6]

Permeability of membrane models

Vesicles of equal surface area contain solute X. Values are initial outward fluxes at the same temperature and the same starting X gradient. Channel Z conducts X when open. The inhibitor is proposed to block Z.

Membrane	Inhibitor	Flux / arbitrary units
Lipid only	Absent	2
Lipid only	Present	2
Lipid + Z	Absent	18
Lipid + Z	Present	6

(a) Calculate the fraction of channel-associated flux removed by the inhibitor, correcting for the lipid-only background. **[2]**

(b) Explain why the results support facilitated diffusion and do not establish active transport. **[2]**

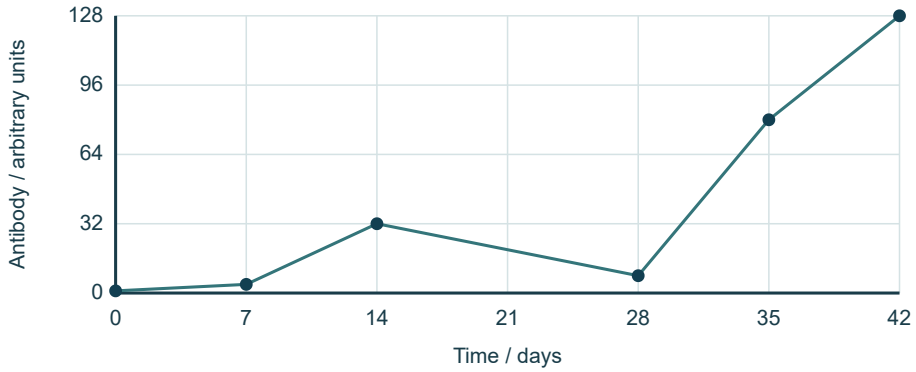
(c) State why the lipid-only inhibitor control is needed and suggest one further measurement of selectivity. **[2]**

QUESTION 3 [6]

Two exposures to an antigen

An individual receives a first dose of antigen on day 0 and a booster on day 28. Antibody values are in arbitrary units. Samples were collected at the times shown.

Day	0	7	14	28	35	42
Antibody	1	4	32	8	80	128



Antibody / arbitrary units plotted against Time / days. Points are measured means or observations as tabulated.

(a) Compare antibody concentration seven days after each exposure using a ratio. **[2]**

(b) Explain the biological basis for the faster secondary response. **[2]**

(c) Evaluate the claim that these measurements prove the booster prevents disease in all recipients. **[2]**

QUESTION 4 [7]

Counting dividing cells

At one instant, observers classified cells in two root-tip regions. They used the same rules and counted complete fields. The mitotic index is the fraction of cells observed in mitosis.

Region	Interphase	Prophase	Metaphase	Anaphase	Telophase
P	144	18	8	4	6
Q	171	9	5	2	3

(a) Calculate the mitotic index in each region. **[2]**

(b) A student says P must have a cell cycle half as long as Q. Evaluate this inference. **[2]**

(c) Describe a way to test inter-observer reliability without counting different fields. **[2]**

(d) Give a visible criterion distinguishing anaphase from metaphase. **[1]**